Question			Answer	Marks	AO	Guidance
12	(a)		Weight = $-2g\mathbf{j}$ N [$=-19.6\mathbf{j}$ N]	B1	2.5	Allow equivalent column vectors in all part questions
				[1]		
12	(b)		horizontal force $3\mathbf{i}$	B1	3.3	May be implied by correct resultant force
			Newton's second law $(-4\mathbf{i}+12\mathbf{j})+3\mathbf{i}-19.6\mathbf{j}=2\mathbf{a}$	M1	1.1a	Must be vector equation. allow one missing or incorrect force.
			$\mathbf{a}=-0.5\mathbf{i}-3.8\mathbf{j}$ m s^{-2}	A1	1.1b	ISW if the magnitude is also given
			Alternative method Horizontal motion $-4+3=2a_x$ Vertical motion $12-19.6=2a_y$	B1 M1		3 N force used in the horizontal equation and not used in the vertical equation Considers motion in two directions. Allow one missing or incorrect force
			$\mathbf{a}=-0.5\mathbf{i}-3.8\mathbf{j}$ m s^{-2}	A1		ISW if the magnitude is also given
				[3]		
12	(c)		$v=\mathbf{u}+\mathbf{a}t$ $=5\mathbf{i}+(-0.5\mathbf{i}-3.8\mathbf{j})\times 4$	M1	1.1a	Using *suvat* equation(s) leading to a vector **v**. Do not award if scalar added to vector
			$=3\mathbf{i}-15.2\mathbf{j}$ m s^{-1}	A1	1.1b	Mark final answer. Must be vector **v** and not speed. FT their vector acceleration
				[2]		
12	(d)		constant velocity is equilibrium so $(\mathbf{i}+7.6\mathbf{j})$ N	B1	1.1b	Cao. ISW if the magnitude is also given
				[1]		